

# Current And Target Flow

## Current system sequence

sequenceDiagram

```
participant J as Jenkins
participant A as FastAPI webhook
participant E as EventStore JSONL
participant R as Regex analyzer
participant G as GitHub service
participant GH as GitHub API

J->>A: POST /api/v1/webhooks/jenkins/failure
A->>A: Normalize payload and console_text
A->>E: Store StoredEvent
A->>R: _analyze_event(event)
R-->>A: Single analysis object
A->>E: Save analysis back to event
A-->>J: 202 Accepted

Note over A,G: Later operator flow
A->>G: prepare-fix / apply-fix
G->>GH: fetch file or push file
GH-->>G: branch/PR result
G-->>A: proposal or push result
A->>E: Persist proposal/github_result
```

## Current component view

flowchart LR

```
Jenkins[Jenkins failure webhook]
App[FastAPI app]
JS[jenkins_service.py]
GS[github_service.py]
```

```
UI[dashboard_ui.py]
Store[runtime/requests.json]
GH[GitHub API]
```

```
Jenkins --> App
App --> JS
App --> GS
App --> UI
JS --> Store
JS --> GS
GS --> GH
```

## Target system sequence

sequenceDiagram

```
participant J as Jenkins
participant A as FastAPI webhook
participant E as EventStore and RunStore
participant LG as LangGraph runner
participant N1 as Node01 extract issues
participant N2 as Node02 fetch and plan
participant N3 as Node03 validate BOM
participant N4 as Node04 push branch
participant GH as GitHub API
```

```
J->>A: POST failure webhook with console_text
A->>E: Persist raw event
A->>LG: Start graph with event_id
A-->>J: 202 Accepted
```

```
LG->>N1: Parse vulnerability output
N1->>E: Save issue inventory
LG->>N2: Fetch manifests and build candidate plans
N2->>GH: Read repo files if needed
GH-->>N2: pom/gradle/docker contents
N2->>E: Save remediation candidates
LG->>N3: Validate BOM-friendly plan
N3->>E: Save validation result
```

```
alt validation passes
```

```
    LG-->>N4: Push approved change set
    N4-->>GH: Create branch, commit files, open PR
    GH-->>N4: branch and PR result
    N4-->>E: Save final status
else manual review required
    N3-->>E: Save blocked or review-required status
end
```

## Target component view

flowchart LR

```
Jenkins[Jenkins]
API[FastAPI webhook and run status APIs]
Store[EventStore and RunStore]
Graph[LangGraph workflow]
N1[Node01 issue extraction]
N2[Node02 manifest fetch and planning]
N3[Node03 BOM validation]
N4[Node04 branch push]
RepoReader[Repo reader service]
Parsers[Manifest parsers]
Validator[Validation service]
Writer[GitHub writer service]
GH[GitHub API]
```

```
Jenkins --> API
API --> Store
API --> Graph
Graph --> N1
Graph --> N2
Graph --> N3
Graph --> N4
N2 --> RepoReader
N2 --> Parsers
N3 --> Validator
N4 --> Writer
RepoReader --> GH
Writer --> GH
N1 --> Store
N2 --> Store
```

N3 --> Store

N4 --> Store